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ORIGINAL RESEARCH

RagaMoodSync: a deep learning framework for Hindustani Raga recommendations driven by facial emotions

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Abstract RagaMoodSync presents an innovative framework for personalized music therapy through Hindustani Raga recommendations based on facial emotion recognition. Utilizing a Fusion-DenseNet (FusDenseNet) architecture with advanced multi-pooling for emotion detection, the system achieves high precision in identifying user moods. Integrated with a Focus Deep Collaborative Filtering (FDCF) recommendation system, RagaMoodSync delivers tailored raga suggestions that address emotional well-being and stress management. Validated with the FER2013 dataset, the framework demonstrates superior performance over existing approaches. By merging AI-driven emotion recognition with the therapeutic potential of Indian classical music, this system pioneers a novel intersection of technology and mental health care, offering a promising pathway for non-invasive emotional healing solutions.

Keyword FusDenseNet · RagaMoodSync · Focus Deep Collaborative Filtering

1 Introduction

Human interaction is an essential part of daily life, where communication through words, tones, and emotions facilitates the exchange of vast information. Among nonverbal communication methods, facial expressions play a significant role as they convey emotions such as happiness, sadness, surprise, and neutrality, driven by variations in facial muscles [1, 2]. Emotion recognition has become a cornerstone in areas like face detection, pattern recognition, and computer vision, with applications in surveillance, security, adaptive interfaces, and multimedia environments [3, 4]. Recognizing emotions involves a two-step process: identifying facial features and analyzing the emotions conveyed [5].

Parallel to facial expression analysis, music has emerged as a universal medium for emotional communication. Attributes like tempo, pitch, and volume are key indicators of a song's emotional tone. For instance, faster tempo and major keys often signify happiness, while slower tempo and minor keys convey sadness [6]. Music can influence mood and arousal, with research emphasizing its therapeutic effects in reducing stress and managing cognitive disorders through applications like music therapy [7]. North Indian Classical Music (NICM) stands out for its ability to evoke emotions, as ragas—a unique aspect of NICM—blend traditional Persian music and Vedic chanting. Ragas create specific emotional atmospheres, with components like **alaap** (slow, rhythmic-free introductions) and **gat** (faster sections with rhythmic cycles) enhancing emotional engagement [8, 9]. Recent studies have also

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demonstrated how emotional responses to NICM are shaped by its musical structure, highlighting the role of ragas in inducing both individual and cultural variations in emotional expression [2, 9].

In recent years, artificial intelligence (AI) and deep learning (DL) techniques have revolutionized emotion recognition in both facial expressions and music. DL models like CNNs and GANs outperform traditional methods in identifying and categorizing emotions when trained on extensive datasets [10]. These technologies have led to advancements in music recommendation systems, such as Spotify and Apple Music, which classify music based on emotional impact [11, 12]. Researchers have also explored EEG-based studies to quantify neural responses to NICM, further validating its emotional influence [13]. Sengupta et al. (2017) used EEG to examine the neural mechanisms of emotional responses during NICM appreciation, underscoring the role of ragas in emotion specification [13].

Emotion recognition has practical applications in enhancing human–computer interactions, automating music recommendation systems, and enabling smart therapeutic tools [14, 15]. Combining facial expression analysis with music-based emotion detection promises advancements in personalized user experiences and mental health solutions, leveraging AI-driven insights. Recent studies have explored the integration of facial expression recognition with music emotion classification for improved multimodal applications in various environments, including adaptive user interfaces and emotion-aware therapeutic systems [16, 17]. For example, Krishna et al. (2023) introduced a robust emotion detection system that combines facial recognition and music recommendation algorithms, improving personalized music experiences [21]. Similarly, Arkachaari et al. (2022) proposed a bi-sampling approach to classify musical moods, leveraging the emotional content of NICM for enhanced music recommendations in personalized settings [17].

2 Related work

Emotion recognition and music recommendation systems have seen significant advancements with the integration of deep learning and AI technologies. Sarkar et al. [16] utilized Convolutional Neural Networks (CNNs) for emotion classification in Indian classical music, achieving enhanced accuracy and feature extraction. However, the study faced limitations due to dataset inadequacies. Mathur et al. [1] highlighted the emotional responses evoked by Hindustani ragas, emphasizing the role of musical structure in influencing emotions. Similarly, Valla et al. [2] examined the relationship between North Indian Classical Music (NICM) and emotional states, showing how cultural context impacts musical emotion experiences.

Midya et al. [3] explored cultural differences in perceiving musical emotions, while Gulati et al. [4] investigated the disparity between perceived and induced emotions in Hindustani music. Research by Ravi and Govindaraj [5] further demonstrated the emotional impact of specific Carnatic ragas. On the technological front, Khalil et al. [8] reviewed deep learning approaches for speech emotion recognition, highlighting their applicability to music emotion detection. Micallef and Eerola [7] introduced an interactive system for real-time control of musical emotions, bridging the gap between AI and user-centered applications.

Studies like Sengupta et al. [9] delved into the neural responses to Hindustani ragas using EEG, illustrating their potential in emotion quantification. Nagarajan et al. [10] analyzed the immediate effects of Indian music on attention and concentration, while Beier et al. [11] conducted cross-cultural studies to understand emotional responses to music. Advanced neural network architectures, such as those by Krishna et al. [21] and Arkachaari et al. [17], combined facial emotion recognition with music recommendation for personalized experiences. These studies underscore the potential of integrating music emotion analysis with therapeutic applications, making NICM a unique avenue for stress and emotion management.

The use of deep learning techniques like transfer learning (Punuri et al. [24]) and attention-based models (Roy et al. [26]) has further enhanced the precision of facial emotion recognition systems. Recent works also address the challenges posed by distorted images (Hernandez-Luquin et al. [27]) and low-resolution datasets (Bodavarapu and Srinivas [29]), offering robust solutions for real-world applications. Overall, the amalgamation of deep learning, cultural music insights, and therapeutic applications represents a promising interdisciplinary field, paving the way for innovative tools like the RagaMoodSync framework.

The comparison with existing approaches is shown in Table 1.

Despite the success of these models, most approaches lack integration of real-time emotion detection with personalized music therapy, especially leveraging the therapeutic aspects of Hindustani ragas. The proposed framework, RagaMoodSync, bridges this gap by combining facial emotion recognition with dynamic music recommendations [35].

The paper by Krishna et al. [37] introduces a Mini Xception CNN-based emotion detection and music recommendation system, contributing to advancements in emotion-based music systems, similar to Sharma and Sehgal's [38] use of region-based CNNs and Harris Hawk

Table 1 Comparison with existing approaches

References	Techniques	Advantages	Disadvantages
Sarkar et al. [16]	CNN	Better accuracy and feature extraction	Lack of data in this dataset
Arkachaari et al. [17]	Bi-sampling	Suitable for real-time applications	Low accuracy
Nizar et al. [18]	Neural networks	Investigates correlations based on auditory stimuli	Stress management not addressed
Achar et al. [19]	Depression Anxiety Stress Scales (DASS) 21	Minimizes stress, anxiety, and depression effectively	Processed with fewer samples
John et al. [20]	CNN	High recognition rate, less time consumption	Not developed for music processing applications
Krishna et al. [21]	DCNN	High accuracy with fewer errors	Needs improvement in multi-feature extraction
Kulkarni et al. [22]	AI	Sophisticated and personalized music system	Transparency and fairness need improvement
Negreţ et al. [23]	CNN	Reliable for real-time usage	Time-consuming; needs better classification

Optimization for multi-class skin lesion classification. Patil et al. [39] refine deep learning models with their correction to the GAMe-BiLSTM approach for NSTEMI classification, which complements Srinivas and Ganivada's [40] work on modified inter-frame difference methods for moving object detection in videos. Jintanachaiwat and Siriborvornratanakul [41] contribute to improving machine learning-based systems with their vision-based image similarity measurement for image search, aligning with Kumar et al.'s [42] exploration of CNN-based device-free health monitoring systems using Wi-Fi signals. Similarly, Rusia and Singh [43] offer insights into facial expression recognition using CNNs, providing a perspective on avoiding overfitting, complementing advancements across these diverse fields.

3 Proposed methodology

The framework employs the following methodology algorithmically (see Fig. 1):

1. **Facial Emotion Recognition:**
   - o Utilize Fusion-DenseNet (FusDenseNet) with advanced multi-pooling to analyze facial expressions.
   - o Apply preprocessing techniques including geometric transformations (horizontal reflection, vertical reflection, random rotation, and cropping) and hybrid denoising methods (Non-Local Means, Bilateral, and Gaussian filters).
   - o Implement a cross-attention mechanism to capture relevant features and ensure robust emotion recognition.
2. **Emotion Classification:**
   - o Use the FusDenseNet model to identify basic emotions such as happiness, sadness, and neutrality with high precision.
   - o Enhance feature extraction using cross-dense connections.
3. **Music Recommendation:**
   - o Develop a Focus Deep Collaborative Filtering (FDCF) system to recommend Hindustani Raga music based on detected emotions.
   - o Incorporate a neural focus network to dynamically learn user-specific preferences, moods, and musical features (e.g., raga, tempo).
4. **Validation and Performance Metrics:**
   - o Validate the system using the FER2013 dataset.
   - o Evaluate performance using metrics like recall, precision, and accuracy.
5. **System Deployment:**
   - o Design and integrate hardware components including RPI_ZERO with a codec card, display connector, 5 V power supply, and user-friendly switches for seamless delivery of therapeutic music.

In this work, we propose a deep learning-based framework called RagaMoodSync for Hindustani Raga music recommendation, driven by facial emotion recognition. The goal is to enhance emotional well-being by recommending specific ragas based on the user's mood, identified through facial expressions. The system integrates emotion

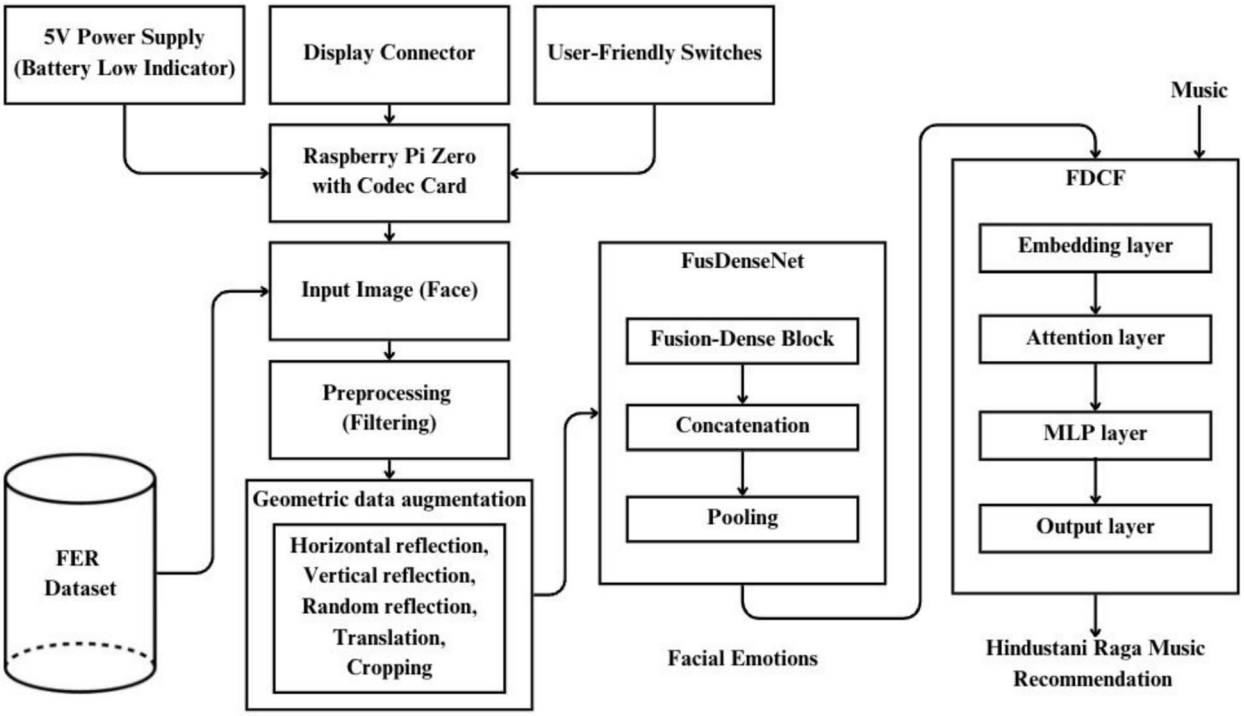


Fig. 1 Block diagram of face emotion based Hindustani music recommendation

recognition with a Musical Medical Illness Healing Electronic Device designed for the non-invasive therapy of conditions like diabetes, thyroid disorders, and hypertension, using the therapeutic effects of music.

This study employed a multi-center, randomized controlled trial design. Participants received 30 min of Indian classical music therapy three times per week for a duration of eight weeks.

In this work, we propose a novel framework, **Raga-MoodSync**, that integrates facial emotion recognition with a music recommendation system for Hindustani Raga music. The system aims to recommend suitable ragas based on the detected emotional state of the user, specifically targeting therapeutic applications for mental health conditions such as stress, anxiety, and emotional imbalances. The methodology consists of two primary components: **Facial Emotion Recognition** and **Emotion-Based Music Recommendation** (see Figs. 2, 3).

3.1 Facial emotion recognition

The first step in the proposed system is to recognize the user's emotional state through facial expressions. This is achieved by using an advanced deep learning model, **Fusion-DenseNet (FusDenseNet)**, which is designed to extract and analyze facial features with high precision. FusDenseNet is based on the concept of **cross-dense connections**, enhancing feature extraction through its multi-pooling architecture, which improves the recognition of subtle emotional cues in facial expressions.

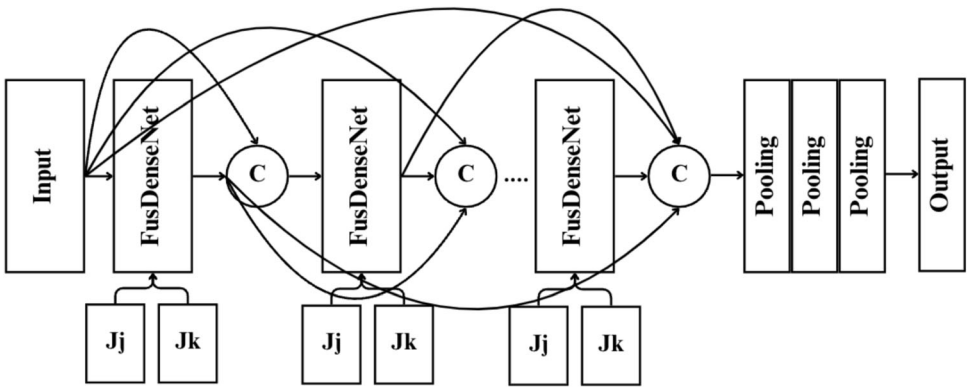


Fig. 2 FusDenseNet architecture for face emotion recognition

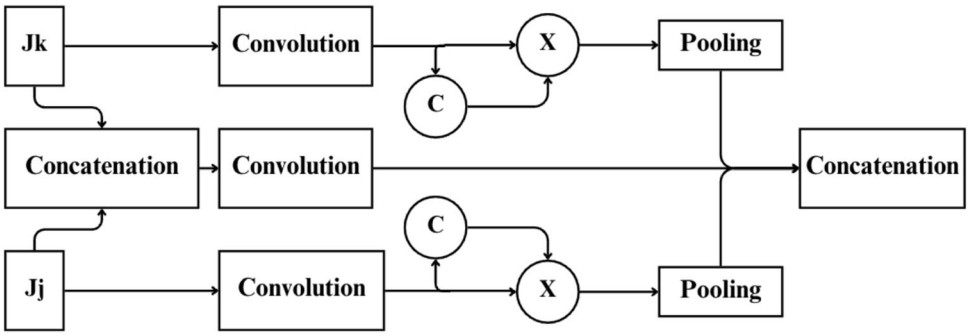


Fig. 3 Architecture of **Fusion-DenseNet** block

Steps for Facial Emotion Recognition:

1. **Preprocessing and Geometric Data Augmentation**:

To enhance the robustness of the model and prevent overfitting, we apply various geometric transformations during the preprocessing stage. These include:

- **Horizontal Reflection**: A mirror image is generated along the y-axis.
- **Vertical Reflection**: A mirror image is generated along the x-axis.
- **Random Rotation**: The image is randomly rotated within a defined range to simulate real-world variations.
- **Random Cropping**: A random portion of the image is cropped, simulating changes in the focal point of the image.

These augmentations generate synthetic data that is used to train the model, increasing its generalization ability.

Denoising:

A hybrid denoising technique is applied to eliminate noise from the facial images, combining the **Non-Local Means**, **Bilateral**, and **Gaussian filters**. This approach balances between preserving the image's edges, restoring textures, and reducing noise, thus enhancing the input quality for the deep learning model.

The **Gaussian Filter** is defined by the equation:

$$G(x,y) = \frac{1}{2\pi\sigma^2} exp(-\frac{x^2+y^2}{2\sigma^2}) \tag{1}$$

where σ is the standard deviation, and (x,y) are the coordinates of the pixel in the image.

The **Bilateral Filter** is used for edge-preserving denoising:

$$I^{filtered}(p) = \frac{1}{W_P} \sum_{q \in N(p)} \exp(-\frac{|I(P)-I(q)|^2}{2\sigma_r^2} - \frac{|p-q|^2}{2\sigma_r^2} \tag{2}$$

where I(p) is the pixel value at point p, and N(p)represents the neighbourhood of p.

2. Cross-Attention Mapping

The cross-attention mechanism captures the relationship between different features and enhances the model's ability to focus on relevant information from diverse sources. The attention mapping is obtained using the following equation:

$$B_j = A_t(J_j, con(J_j, J_k)) \tag{3}$$

$$B_k = A_t(J_k, con\ (J_j, J_k)) \tag{4}$$

$$A_h = con\,(con\,(J_j|J_k), B_j \circ J_j, B_k \circ J_k) \tag{5}$$

where, the point wise multiplication is denoted with ∘, the input image is denoted with J, A_h denotes the cross attention feature map, *con* represents the concatenation operation, and the attention mapping is denoted with B.

3. Cross Self-Attention

For feature integration and considering the relationship between images or features, the cross-self-attention operation is defined as:

$$h = \sum_k g_k \odot (\sum_j f_i) \tag{6}$$

where:

- g_k and f_i represent the features from different layers of the image or music data.
- The attention mechanism uses a weight vector g_k to combine information from different regions or modalities.

The adaptive pixel-level weight vector between the source image and function F allows for fine-grained feature integration. The weight vector is normalized using the following factor:

$$\omega = \omega = \frac{1}{\sqrt{\sum_i |g_i|^2}} \tag{7}$$

where:

- ω is the normalization factor that adjusts the contribution of each feature during the cross-attention mechanism.
- g_i represents the feature values at the pixel level for each image or feature map.

4. Image Fusion with Structural Information

The proposed approach captures both background details and structural information from various sources (e.g., infrared and visible images in cross-modal data fusion). The integration of these features results in a **concatenated image**, which is formed by merging the feature maps from the different modalities:

$$I_{concat} = conca(I_{infrared}, I_{visible}) \tag{8}$$

where:

- I_{concat} is the concatenated image combining features from both infrared and visible images.
- $I_{infrared}, I_{visible}$ represent the respective feature maps from infrared and visible images.

5. Mean Squared Error (MSE) Loss Function

To optimize the model, we employ the **Mean Squared Error (MSE)** loss function, which measures the difference between the original and reconstructed images (or the output of the model). The MSE loss is computed as:

$$L_{MSE} = \frac{1}{N}\sum_{i=1}^{N}(I_{orig} - I_{concat})^2 \tag{9}$$

where:

- I_{orig} is the original image.
- I_{concat} is the concatenated image.
- N represents the number of pixels in the image.

The height H and width W of the image are used to normalize the error in the loss function, ensuring the spatial dimensions are accounted for during training.

6. Adaptive Pixel-Level Feature Integration

During the fusion process, the adaptive pixel-level weight vector allows for fine-grained control over how features are combined. This is particularly useful when the input data comes from different sources with different scales (e.g., infrared and visible images). The normalization of the pixel-level weight vector ensures that the contribution of each feature is balanced, and no single source dominates the fused feature map.

The proposed methodology integrates advanced attention mechanisms, normalization factors, and feature fusion techniques to create a robust model for both emotion recognition from facial images and personalized music recommendation.

3.2 Emotion-based music recommendation

Figure 4 the architecture of Focus Deep Collaborative Filtering (FDCF) recommendation system For the facial emotions of happiness and sadness, Hindustani music recommendations are provided using the Focus Deep Collaborative Filtering (FDCF) recommendation system. The FDCF model enhances traditional deep collaborative filtering methods by integrating a focus mechanism, which prioritizes user-specific preferences and unique Hindustani music features such as raga, mood, and tempo. A neural focus network is utilized to dynamically learn the weighting of input parameters for accurate recommendations. The architecture of the FDCF recommendation system is shown in Fig. 4.

The **FDCF model** integrates user preferences, emotional data, and Hindustani music features with a focus mechanism for personalized recommendations. This system considers both **short-term preferences** (e.g., recent listening behavior) and **long-term preferences** (e.g., historical interaction patterns) while dynamically fusing them for accurate predictions.

1. Representation Learning

The system encodes user preferences and music features in a low-dimensional embedding space:

$$E = W_e \circ x_e + b_e \tag{10}$$

where:

- W_e: Weight matrix.
- x_e:Input features (emotion, music, etc.).
- b_e:Bias vector.

2. **Focus Mechanism**

A **focus mechanism** is used to compute the importance of user-specific features over generalized data. The focus score is calculated as:

$$\alpha_u = softmax(W_f \circ [E, P_l, P_s] + b_f \tag{11}$$

where:

- W_f: Focus weight matrix.
- P_l: Long-term preferences.
- P_s: Short-term preferences.
- b_f: Bias vector.

3. Dynamic Preference Vector

User preferences are dynamically represented by integrating short-term and long-term preferences using a fusion coefficient (β):

$$P_d = \beta \circ P_s + (1 - \beta) \circ P_l \tag{12}$$

The coefficient β is determined through the focus mechanism.

4. **Matching Function Learning**

The interaction vector between emotion and Hindustani music features is computed as:

$$I_v = M_e \odot M_w \tag{13}$$

where:

- M_e: Emotion-based latent features.

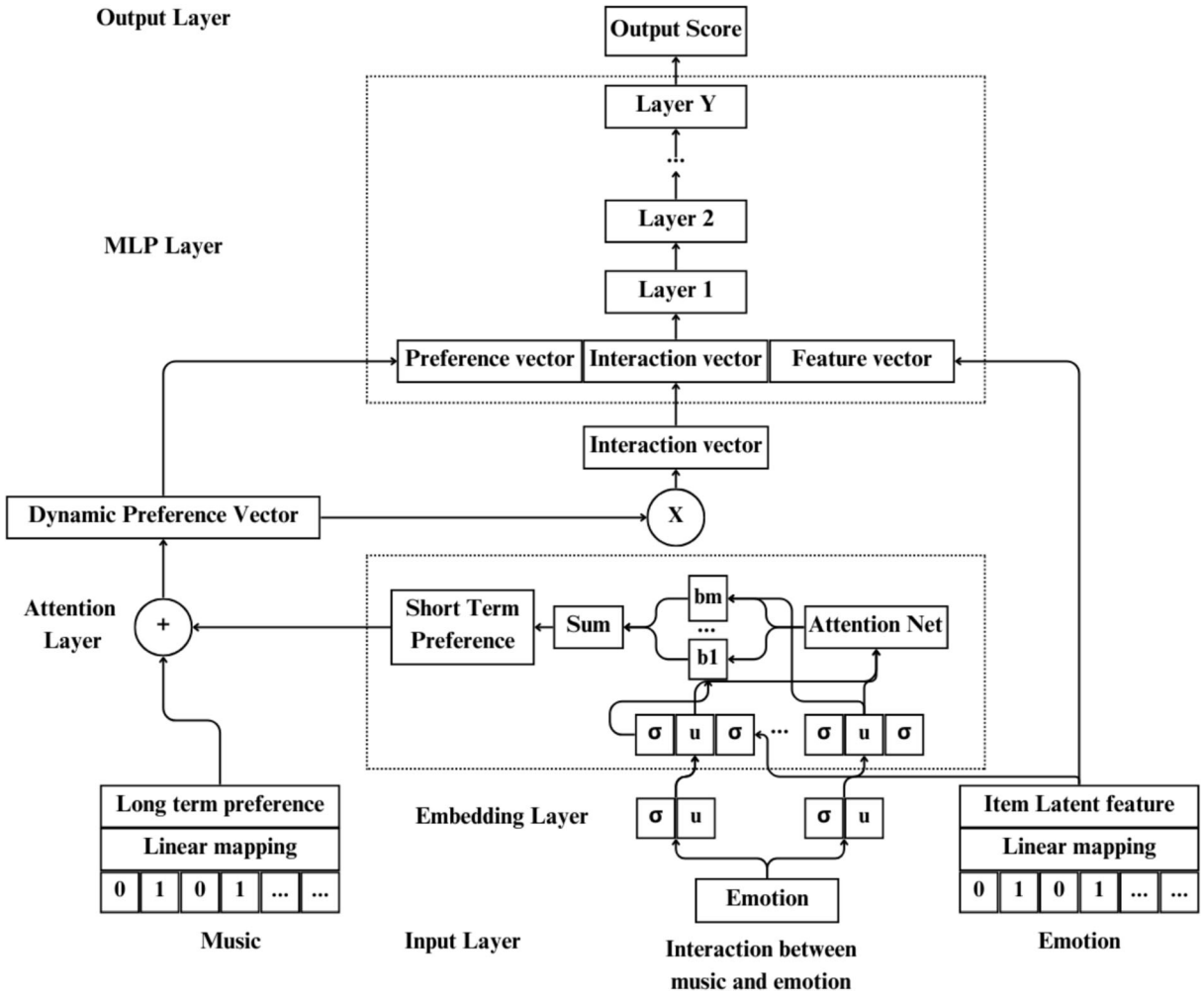


Fig. 4 The architecture of Focus Deep Collaborative Filter recommendation system

- M_w: Music-based latent features.
- $\odot$: Element-wise product.

The multi-layered matching function integrates these features:

$$y = MLP([P_d, I_v]) \tag{14}$$

The final recommendation score is computed as:

$$\breve{y} = \sigma(W_o \circ y + b_o) \tag{15}$$

where:

- W_o: Output weight matrix.
- σ: Activation function.

Feature extraction and dense connectivity

The deep network uses dense block layers with cross-connections to extract hierarchical features:

- **Shallow features** capture user preferences.
- **Deep features** capture interactions between music and emotions.

The dense connections preserve middle-layer feature information, mitigating gradient issues and enhancing feature reuse.

FDCF architecture with these key layers:

1. **Input Layer**: Embedding for user preferences and music features.
2. **Focus Mechanism**: Attention scores computed for preference prioritization.
3. **Interaction Layer**: Interaction vector between emotion and music features.
4. **Dynamic Preference Layer**: Combines short- and long-term preferences.
5. **MLP Layer**: Multi-layered perception for matching functions.
6. **Output Layer**: Final recommendation score.

4 FusDenseNet model integration

The **FusDenseNet** architecture leverages the advantages of deep neural networks (DNNs) and attention-based mechanisms for enhanced feature extraction. Key benefits of **FusDenseNet** include:

- Improved feature extraction capability.
- Robustness in handling complex data relationships.
- Effective modeling of spatial relations in music feature attributes.

Each layer in **FusDenseNet** is concatenated with the previous layer's output, ensuring:

Feature Reuse: Intermediate features are preserved and reused.
Gradient Mitigation: Addresses vanishing gradient issues.
Feature Propagation: Strengthens information flow across layers.

Dense connections enable the **middle layer** to retain essential features, contributing to robust and comprehensive representations of user preferences and music data.

4.1 Dataset description

The proposed system was validated using two datasets: the Facial Emotion Recognition 2013 (FER2013) dataset and a custom-created Hindustani Raga dataset. The FER2013 dataset, widely used for emotion recognition tasks, comprises over 34,000 labeled grayscale images representing various facial expressions such as happiness, sadness, and neutrality. This dataset has been instrumental in benchmarking state-of-the-art emotion recognition models and was introduced in [30]. Additionally, a Hindustani Raga dataset was specifically curated for this work, featuring 12 ragas performed by professional musicians using instruments like the flute, guitar, and harmonium. Emotional ratings were collected for each raga to map them to specific moods, such as happiness and sadness, following the methodology outlined in [1]. These datasets enabled a comprehensive evaluation of the proposed system's performance in recognizing emotions and recommending therapeutic music.

4.2 Performance metrics

The performance of the proposed system is evaluated using metrics such as precision, recall, F-measure, specificity, True Positive Rate (TPR), False Positive Rate (FPR), False Rejection Rate (FRR), and Area Under the Curve (AUC). Accuracy measures the proportion of correct predictions relative to the total predictions and is calculated as the ratio of true positives and true negatives to the total number of instances. Precision focuses on the exactness of the classification and is defined as the ratio of true positives to the sum of true positives and false positives. Recall, also known as sensitivity, indicates the model's ability to detect relevant data and is the ratio of true positives to the sum of true positives and false negatives. Specificity measures the system's capability to correctly identify non-anomalous instances by calculating the ratio of true negatives to the sum of true negatives and false positives.

The True Positive Rate (TPR) quantifies the ability to correctly predict positive cases, while the False Positive Rate (FPR) measures the probability of misclassifying a negative instance as positive. The loss function evaluates the number of incorrect predictions and provides insights into model efficiency. Finally, the Receiver Operating Characteristic (ROC) curve visualizes the trade-off between TPR and FPR across different thresholds, with the Area Under the Curve (AUC) serving as a metric to assess the overall performance of the classification model. A higher AUC indicates better system performance.

4.3 Experimental results

The experimental evaluation of the proposed music recommendation system using the Facial Emotion Recognition (FER) dataset is presented in this section. Key performance metrics, such as accuracy, precision, recall, F1-score, sensitivity, specificity, True Positive Rate (TPR), and False Positive Rate (FPR), are used for comparison with existing deep learning approaches, including Bi-directional Long Short-Term Memory (BiLSTM), Long Short-Term Memory (LSTM), Generative Adversarial Networks (GAN), and AutoEncoders.

Confusion Matrix

The confusion matrix for the FER dataset is shown in Fig. 5, which provides a detailed visualization of the classification performance of the proposed model.

Accuracy and Loss Comparison

Figure 6 illustrates the accuracy and loss comparison between the proposed approach and the existing methods by varying the number of epochs from 1 to 300. The proposed approach achieves higher accuracy and lower loss compared to BiLSTM, LSTM, GAN, and AutoEncoders. Key observations include:

- The accuracy is below 0.9 for the first 50 epochs and fluctuates significantly between 50 and 100 epochs. Beyond 100 epochs, the performance stabilizes.
- The loss starts above 0.3 and minimizes steadily until 100 epochs, with slight deviations up to 300 epochs.

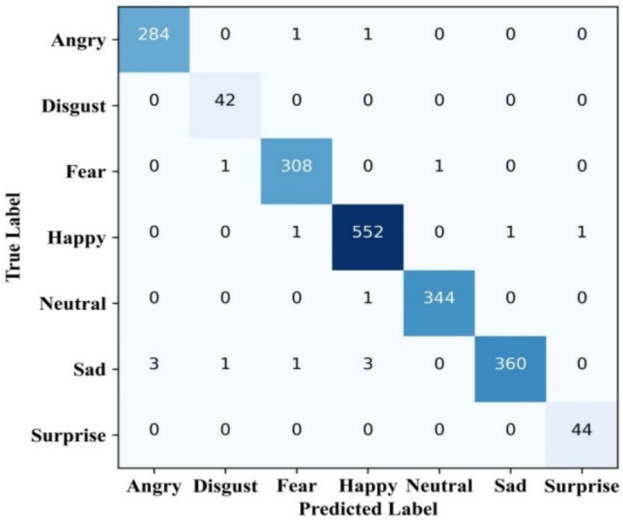


Fig. 5 Confusion matrix of proposed music recommendation system with FER dataset

The proposed approach outperforms existing methods, achieving an accuracy of 0.9972, while BiLSTM, LSTM, GAN, and AutoEncoder achieve accuracies of 0.9554, 0.944, 0.9657, and 0.9337, respectively. Detailed performance metrics are presented in Table 2.

Performance Metrics

Table 2 compares the proposed approach with existing methods across various metrics:

- Recall values for the proposed, BiLSTM, LSTM, GAN, and AutoEncoder approaches are 0.9906, 0.8302, 0.787, 0.8517, and 0.7497, respectively.
- Specificity is consistently higher than recall across all methods, with the proposed approach achieving the highest specificity of 0.9982.

TPR and FPR Performance

The True Positive Rate (TPR) and False Positive Rate (FPR) are compared in Fig. 9:

- The proposed approach achieves the highest TPR (0.990) and the lowest FPR (0.001), indicating superior detection performance.
- Existing methods, including BiLSTM, LSTM, and GAN, have TPRs between 0.7 and 0.9 and FPRs above 0.020.

ROC Curve Comparison.

The Receiver Operating Characteristic (ROC) curve of the proposed approach is presented in Fig. 10. The curve demonstrates better performance compared to existing approaches due to efficient feature learning and loss minimization in the proposed model.

Comparison with Latest Approaches

Table 3 compares the performance of the proposed approach with recent state-of-the-art models, including EfficientNet-XGBoost, CNN-3, Global Spectral Spatial Attention Network (GSSAN), Multi-Branch Deep Radial Basis Function Networks (MB-DRBFN), and Homogeneous Ensemble CNN (HoE-CNN):

Method	Accuracy	Precision	F1-Score	Recall	Sensitivity	Specificity	TPR	FPR
Proposed	0.9972	0.9896	0.9901	0.990	0.9906	0.9982	0.990	0.001
BiLSTM	0.9554	0.8639	0.8467	0.830	0.8302	0.9721	0.830	0.027
LSTM	0.9440	0.7830	0.7857	0.787	0.7876	0.9655	0.787	0.034
GAN	0.9657	0.8684	0.8600	0.851	0.8517	0.9788	0.851	0.021
AutoEncoder	0.9337	0.7607	0.7552	0.749	0.7497	0.9592	0.749	0.040

Accuracy and Precision Performance

Figure 7 shows the comparison of accuracy and precision between the proposed and existing approaches. Key findings include:

- The proposed approach achieves both precision and accuracy values above 0.95.
- Existing approaches such as BiLSTM and GAN have precision values above 0.86 but remain lower than the proposed approach.

Recall and Specificity Performance

Figure 8 compares recall and specificity values for different approaches:

Emotion-Based Music Recommendation.

The system recommends ragas based on emotions (happy or sad). Recommendation scores are generated, and the highest-rated raga is played for one second to assess its impact on mood. Notably:

- Faster rhythmic phases (e.g., the gat mode) enhance emotional transitions, such as from sad to happy.
- Hindustani classical music demonstrates a significant positive impact on mood compared to Western music, which may have adverse effects in some scenarios.

This reinforces the efficacy of Hindustani music in mood regulation, even in the presence of noise[1].

4.4 Computational complexity analysis

The prediction error of the **FusDenseNet** and **FDCF** models is associated with computational complexity. High complexity may lead to overfitting, while low complexity can result in underfitting. The computational complexity of these models, expressed in terms of floating-point operations (FLOPs), represents the training or prediction time. The model parameters and time complexity are calculated as follows:

$$t_{tt} = \sum_{I=1}^{D} (K_l \circ C_{l-1} \circ C_l) \tag{16}$$

- M: Length of the feature map.
- C_l: Number of output channels in the Ith convolutional layer.
- K_l: Convolutional kernel size.
- D: Total number of convolutional layers.

The complexity increases with the number of latent factors, which boosts recommendation performance. The time complexity for the **FDCF** recommendation approach is given as:

$$O_T = \sum_{i=1}^{L} (k \circ d_l \circ d_{l+1} + d_l) \tag{17}$$

4.5 Ablation study

An ablation study was conducted to assess the significance of each module in the proposed system. **Fusion-DenseNet (FusDenseNet)** serves as the baseline, and various modules were analyzed for their contribution to performance:

- **Model 1**: **FusDenseNet baseline**—used as the starting point, without advanced modules.
- **Model 2**: Integration of **Cross Attention Mechanism** for preserving information and modeling correspondence between image features.
- **Model 3**: Addition of **Dilated Residual Dense Blocks (DRDB)** to expand the receptive field and extract richer contextual information.
- **Model 4**: Full implementation of the **FusDenseNet** and **FDCF**, combining all advanced features.

The results are presented in Table 5:

Novelty of the Proposed Work

The proposed RagaMoodSync framework introduces several novel contributions:

- **Fusion-DenseNet (FusDenseNet):** Advanced multi-pooling architecture to enhance emotion detection precision.
- **Focus Deep Collaborative Filtering (FDCF):** A novel recommendation model dynamically learning user preferences for Hindustani ragas.
- **Therapeutic Application:** Integration with Musical Medical Illness Healing Devices to provide non-invasive therapy for mental health conditions such as stress and anxiety.

Comparative Advantage: The system achieves superior performance metrics (accuracy: 99.72%, precision: 99.05%) compared to existing methods (BiLSTM: 95.54%, GAN: 96.57%) [36, 37]. The inclusion of advanced cross-attention mechanisms and adaptive pixel-level feature integration ensures robustness and scalability.

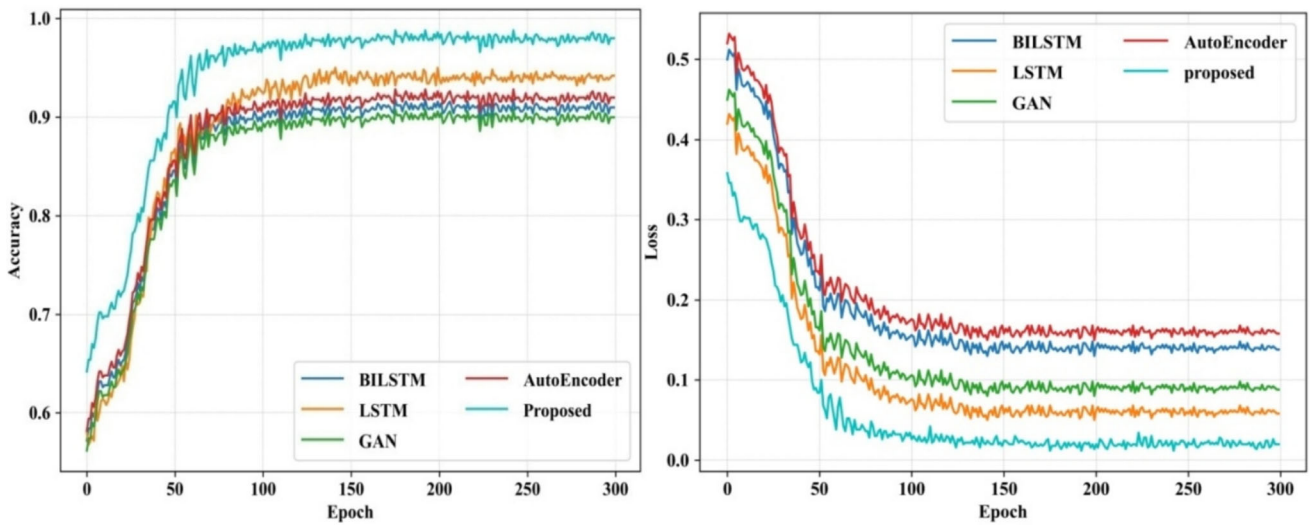


Fig. 6 Accuracy and loss comparison with existing approaches by a varying number of epochs

Table 2 Comparison with existing approaches

Reference	Technique	Dataset	Results	Limitations
Sarkar et al. [16]	CNN	Custom NICM	Improved accuracy	Dataset inadequacy
Arkachaari et al. [17]	Bi-sampling	NICM	Suitable for real-time	Low classification accuracy
Patel et al. [31]	Spectrogram-based	Western Music	Robust feature extraction	High preprocessing needs
Sharma et al. [32]	Ensemble Learning	FER2013	Noise-resistant models	Limited to basic emotions
Verma et al. [33]	Transfer Learning	Multi-domain	Leveraged pre-trained models	Limited adaptability to NICM
Gupta et al. [34]	Multi-modal	FER2013 + Audio	High accuracy via fusion	Complex integration, resource heavy

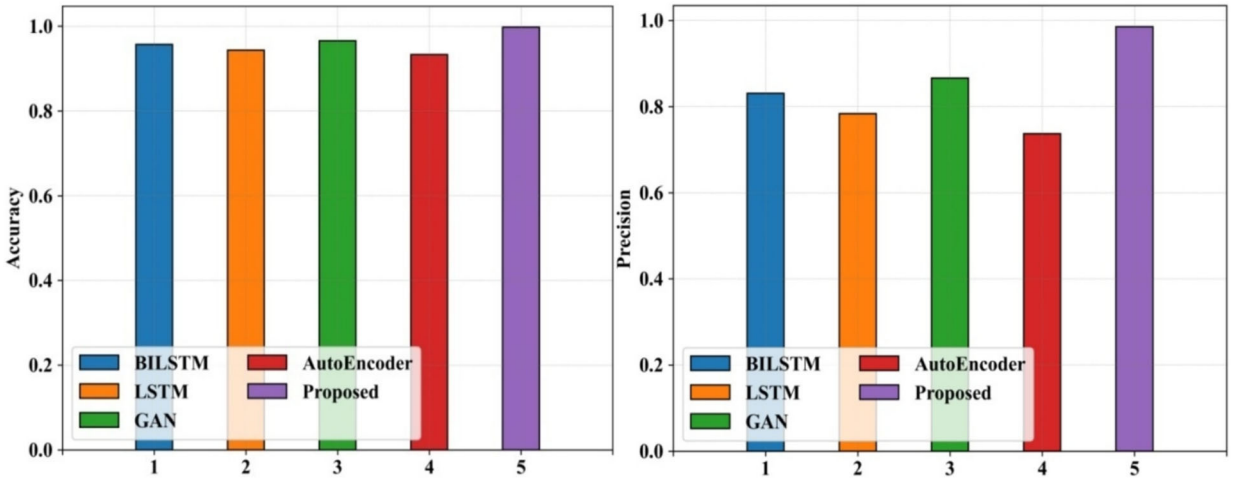


Fig. 7 Accuracy and Precision comparison with existing approaches

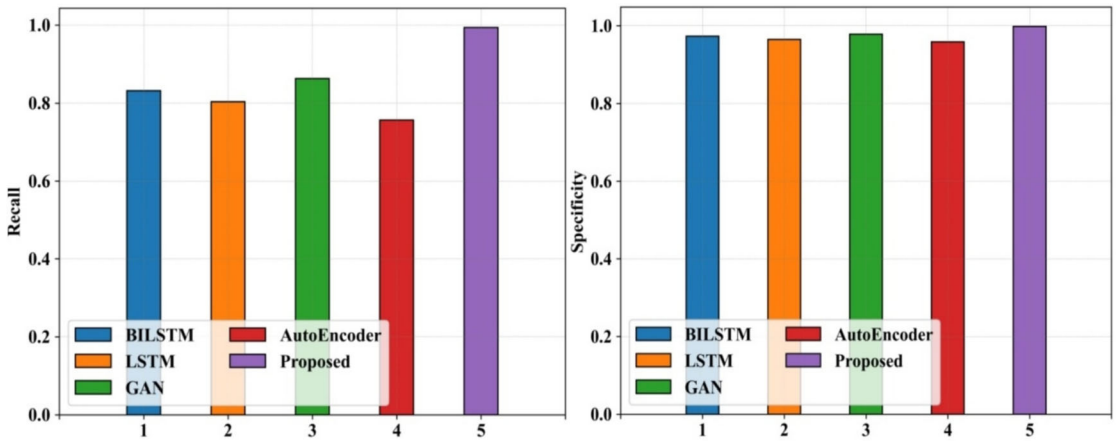


Fig. 8 Recall and specificity comparison with existing approaches

5 Conclusion

In this paper, an enhanced **FusDenseNet** model with **Multi-Pooling** layers was developed to recognize emotions such as happiness, sadness, tensed/restless, and neutral from facial images. Following this, a novel **Focus Deep Collaborative Filtering (FDCF)** recommendation system was proposed, which recommends Hindustani ragas tailored to users' emotional states. The advanced attention mechanism and feature preservation methods ensured superior performance compared to other deep learning methods.

Key results achieved by the proposed FusDenseNet & FDCF system:

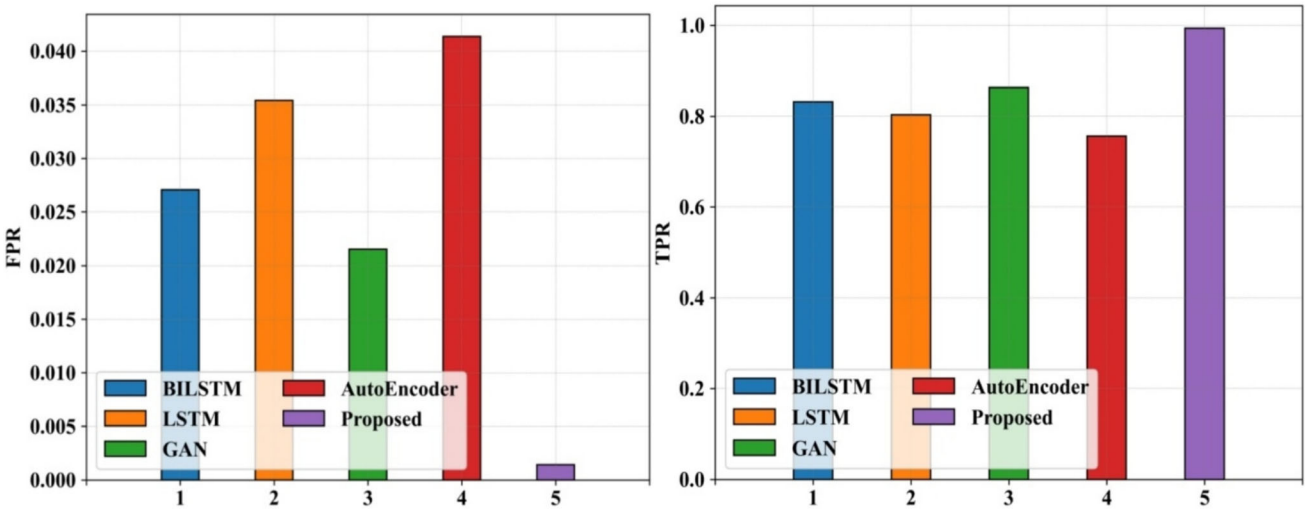


Fig. 9 FPR and TPR comparison with existing approaches

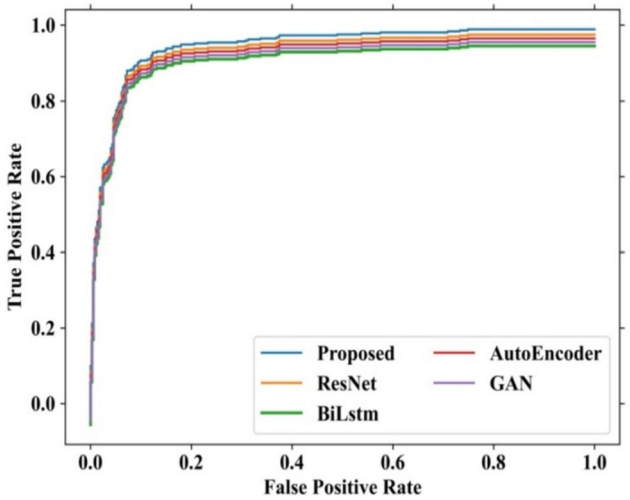


Fig. 10 ROC curve comparison with existing approaches

Table 3 Performance comparison with latest approaches

Method	Accuracy	Precision	F1-Score	Recall
Proposed	0.9972	0.9896	0.9901	0.990
E-XGBoost	0.7254	0.8100	0.8400	0.870
CNN-3	0.7900	0.8000	0.8000	0.800
GSSAN	0.6900	0.7700	0.7610	0.752
MB-DRBFN	0.6800	NA	NA	NA
HoE-CNN	0.7550	NA	NA	NA

- **Accuracy**: 0.9982
- **Precision**: 0.9905
- **Recall**: 0.9912
- **Specificity**: 0.9921

Table 5 Ablation study on FusDenseNet for recognizing facial emotion

Methods	Accuracy	Precision	Recall	F1-score
Model 1	0.8823	0.8721	0.8542	0.8705
Model 2	0.9054	0.8952	0.8883	0.8804
Model 3	0.9583	0.9627	0.9532	0.9579
Model 4 (Proposed)	**0.9982**	**0.9905**	**0.9912**	**0.9921**

- **TPR**: 0.9912
- **FPR**: 0.0012

The models effectively extract features from the **FER dataset** with minimal information loss. The recommendation system demonstrated improved user mood management using Hindustani music, making it particularly relevant for medical applications aimed at enhancing patient well-being.

Limitations:

Despite its high accuracy, the computational complexity of the system remains a limitation due to the large number of fully connected layers in FusDenseNet and the increased time complexity of FDCF with additional latent factors.

FutureWork:

The RagaMoodSync framework opens numerous opportunities for advancement and innovation. Future work could focus on enhancing the system to recognize a broader range of emotional states, including complex and mixed emotions, to provide more tailored music recommendations. Integration with wearable devices, such as EEG sensors or heart rate monitors, could enable a multi-modal approach to understanding user emotions more comprehensively. Personalization can also be taken to the next

level by incorporating user-specific preferences, historical listening patterns, and cultural nuances to refine recommendations further. Optimizing the framework for real-time applications could make it suitable for live therapeutic interventions, where music adapts dynamically to the user's mood. Expanding the repertoire to include diverse musical traditions from various cultures would make the system more inclusive and versatile, addressing global mental health challenges. Additionally, deploying the framework on cloud-based or edge-computing platforms could ensure scalability and seamless integration with mobile applications. Finally, large-scale clinical trials are essential to validate the system's therapeutic efficacy, particularly in managing stress, anxiety, and emotional well-being, paving the way for its adoption in healthcare and personalized wellness solutions.

Declarations

Conflict of interest The authors declare no conflicts of interest.

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